



AP[®] Chemistry Practice Exam

From the 2012 Administration

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PERIODIC TABLE OF THE ELEMENTS

1 H 1.008																	2 He 4.00
3 Li 6.94	4 Be 9.01											5 B 10.81	6 C 12.01	7 N 14.01	8 O 16.00	9 F 19.00	10 Ne 20.18
11 Na 22.99	12 Mg 24.30											13 Al 26.98	14 Si 28.09	15 P 30.97	16 S 32.06	17 Cl 35.45	18 Ar 39.95
19 K 39.10	20 Ca 40.08	21 Sc 44.96	22 Ti 47.90	23 V 50.94	24 Cr 52.00	25 Mn 54.94	26 Fe 55.85	27 Co 58.93	28 Ni 58.69	29 Cu 63.55	30 Zn 65.39	31 Ga 69.72	32 Ge 72.59	33 As 74.92	34 Se 78.96	35 Br 79.90	36 Kr 83.80
37 Rb 85.47	38 Sr 87.62	39 Y 88.91	40 Zr 91.22	41 Nb 92.91	42 Mo 95.94	43 Tc (98)	44 Ru 101.1	45 Rh 102.91	46 Pd 106.42	47 Ag 107.87	48 Cd 112.41	49 In 114.82	50 Sn 118.71	51 Sb 121.75	52 Te 127.60	53 I 126.91	54 Xe 131.29
55 Cs 132.91	56 Ba 137.33	57 *La 138.91	72 Hf 178.49	73 Ta 180.95	74 W 183.85	75 Re 186.21	76 Os 190.2	77 Ir 192.2	78 Pt 195.08	79 Au 196.97	80 Hg 200.59	81 Tl 204.38	82 Pb 207.2	83 Bi 208.98	84 Po (209)	85 At (210)	86 Rn (222)
87 Fr (223)	88 Ra 226.02	89 †Ac 227.03	104 Rf (261)	105 Db (262)	106 Sg (266)	107 Bh (264)	108 Hs (277)	109 Mt (268)	110 Ds (271)	111 Rg (272)							

*Lanthanide Series

58 Ce 140.12	59 Pr 140.91	60 Nd 144.24	61 Pm (145)	62 Sm 150.4	63 Eu 151.97	64 Gd 157.25	65 Tb 158.93	66 Dy 162.50	67 Ho 164.93	68 Er 167.26	69 Tm 168.93	70 Yb 173.04	71 Lu 174.97
90 Th 232.04	91 Pa 231.04	92 U 238.03	93 Np (237)	94 Pu (244)	95 Am (243)	96 Cm (247)	97 Bk (247)	98 Cf (251)	99 Es (252)	100 Fm (257)	101 Md (258)	102 No (259)	103 Lr (262)

†Actinide Series

STANDARD REDUCTION POTENTIALS IN AQUEOUS SOLUTION AT 25°C

Half-reaction			$E^\circ(\text{V})$
$\text{F}_2(\text{g}) + 2\text{e}^-$	$\rightarrow$	2F^-	2.87
$\text{Co}^{3+} + \text{e}^-$	$\rightarrow$	Co^{2+}	1.82
$\text{Au}^{3+} + 3\text{e}^-$	$\rightarrow$	$\text{Au}(\text{s})$	1.50
$\text{Cl}_2(\text{g}) + 2\text{e}^-$	$\rightarrow$	2Cl^-	1.36
$\text{O}_2(\text{g}) + 4\text{H}^+ + 4\text{e}^-$	$\rightarrow$	$2\text{H}_2\text{O}(\text{l})$	1.23
$\text{Br}_2(\text{l}) + 2\text{e}^-$	$\rightarrow$	2Br^-	1.07
$2\text{Hg}^{2+} + 2\text{e}^-$	$\rightarrow$	Hg_2^{2+}	0.92
$\text{Hg}^{2+} + 2\text{e}^-$	$\rightarrow$	$\text{Hg}(\text{l})$	0.85
$\text{Ag}^+ + \text{e}^-$	$\rightarrow$	$\text{Ag}(\text{s})$	0.80
$\text{Hg}_2^{2+} + 2\text{e}^-$	$\rightarrow$	$2\text{Hg}(\text{l})$	0.79
$\text{Fe}^{3+} + \text{e}^-$	$\rightarrow$	Fe^{2+}	0.77
$\text{I}_2(\text{s}) + 2\text{e}^-$	$\rightarrow$	2I^-	0.53
$\text{Cu}^+ + \text{e}^-$	$\rightarrow$	$\text{Cu}(\text{s})$	0.52
$\text{Cu}^{2+} + 2\text{e}^-$	$\rightarrow$	$\text{Cu}(\text{s})$	0.34
$\text{Cu}^{2+} + \text{e}^-$	$\rightarrow$	Cu^+	0.15
$\text{Sn}^{4+} + 2\text{e}^-$	$\rightarrow$	Sn^{2+}	0.15
$\text{S}(\text{s}) + 2\text{H}^+ + 2\text{e}^-$	$\rightarrow$	$\text{H}_2\text{S}(\text{g})$	0.14
$2\text{H}^+ + 2\text{e}^-$	$\rightarrow$	$\text{H}_2(\text{g})$	0.00
$\text{Pb}^{2+} + 2\text{e}^-$	$\rightarrow$	$\text{Pb}(\text{s})$	-0.13
$\text{Sn}^{2+} + 2\text{e}^-$	$\rightarrow$	$\text{Sn}(\text{s})$	-0.14
$\text{Ni}^{2+} + 2\text{e}^-$	$\rightarrow$	$\text{Ni}(\text{s})$	-0.25
$\text{Co}^{2+} + 2\text{e}^-$	$\rightarrow$	$\text{Co}(\text{s})$	-0.28
$\text{Cd}^{2+} + 2\text{e}^-$	$\rightarrow$	$\text{Cd}(\text{s})$	-0.40
$\text{Cr}^{3+} + \text{e}^-$	$\rightarrow$	Cr^{2+}	-0.41
$\text{Fe}^{2+} + 2\text{e}^-$	$\rightarrow$	$\text{Fe}(\text{s})$	-0.44
$\text{Cr}^{3+} + 3\text{e}^-$	$\rightarrow$	$\text{Cr}(\text{s})$	-0.74
$\text{Zn}^{2+} + 2\text{e}^-$	$\rightarrow$	$\text{Zn}(\text{s})$	-0.76
$2\text{H}_2\text{O}(\text{l}) + 2\text{e}^-$	$\rightarrow$	$\text{H}_2(\text{g}) + 2\text{OH}^-$	-0.83
$\text{Mn}^{2+} + 2\text{e}^-$	$\rightarrow$	$\text{Mn}(\text{s})$	-1.18
$\text{Al}^{3+} + 3\text{e}^-$	$\rightarrow$	$\text{Al}(\text{s})$	-1.66
$\text{Be}^{2+} + 2\text{e}^-$	$\rightarrow$	$\text{Be}(\text{s})$	-1.70
$\text{Mg}^{2+} + 2\text{e}^-$	$\rightarrow$	$\text{Mg}(\text{s})$	-2.37
$\text{Na}^+ + \text{e}^-$	$\rightarrow$	$\text{Na}(\text{s})$	-2.71
$\text{Ca}^{2+} + 2\text{e}^-$	$\rightarrow$	$\text{Ca}(\text{s})$	-2.87
$\text{Sr}^{2+} + 2\text{e}^-$	$\rightarrow$	$\text{Sr}(\text{s})$	-2.89
$\text{Ba}^{2+} + 2\text{e}^-$	$\rightarrow$	$\text{Ba}(\text{s})$	-2.90
$\text{Rb}^+ + \text{e}^-$	$\rightarrow$	$\text{Rb}(\text{s})$	-2.92
$\text{K}^+ + \text{e}^-$	$\rightarrow$	$\text{K}(\text{s})$	-2.92
$\text{Cs}^+ + \text{e}^-$	$\rightarrow$	$\text{Cs}(\text{s})$	-2.92
$\text{Li}^+ + \text{e}^-$	$\rightarrow$	$\text{Li}(\text{s})$	-3.05

ADVANCED PLACEMENT CHEMISTRY EQUATIONS AND CONSTANTS

ATOMIC STRUCTURE

$$E = h\nu \quad c = \lambda\nu$$

$$\lambda = \frac{h}{mv} \quad p = mv$$

$$E_n = \frac{-2.178 \times 10^{-18}}{n^2} \text{ joule}$$

EQUILIBRIUM

$$K_a = \frac{[\text{H}^+][\text{A}^-]}{[\text{HA}]}$$

$$K_b = \frac{[\text{OH}^-][\text{HB}^+]}{[\text{B}]}$$

$$K_w = [\text{OH}^-][\text{H}^+] = 1.0 \times 10^{-14} \text{ @ } 25^\circ\text{C}$$

$$= K_a \times K_b$$

$$\text{pH} = -\log [\text{H}^+], \text{pOH} = -\log [\text{OH}^-]$$

$$14 = \text{pH} + \text{pOH}$$

$$\text{pH} = \text{p}K_a + \log \frac{[\text{A}^-]}{[\text{HA}]}$$

$$\text{pOH} = \text{p}K_b + \log \frac{[\text{HB}^+]}{[\text{B}]}$$

$$\text{p}K_a = -\log K_a, \text{p}K_b = -\log K_b$$

$$K_p = K_c(RT)^{\Delta n},$$

where Δn = moles product gas – moles reactant gas

THERMOCHEMISTRY/KINETICS

$$\Delta S^\circ = \sum S^\circ \text{ products} - \sum S^\circ \text{ reactants}$$

$$\Delta H^\circ = \sum \Delta H_f^\circ \text{ products} - \sum \Delta H_f^\circ \text{ reactants}$$

$$\Delta G^\circ = \sum \Delta G_f^\circ \text{ products} - \sum \Delta G_f^\circ \text{ reactants}$$

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$= -RT \ln K = -2.303 RT \log K$$

$$= -n\mathcal{F}E^\circ$$

$$\Delta G = \Delta G^\circ + RT \ln Q = \Delta G^\circ + 2.303 RT \log Q$$

$$q = mc\Delta T$$

$$C_p = \frac{\Delta H}{\Delta T}$$

$$\ln[A]_t - \ln[A]_0 = -kt$$

$$\frac{1}{[A]_t} - \frac{1}{[A]_0} = kt$$

$$\ln k = \frac{-E_a}{R} \left(\frac{1}{T} \right) + \ln A$$

$$E = \text{energy} \quad v = \text{velocity}$$

$$\nu = \text{frequency} \quad n = \text{principal quantum number}$$

$$\lambda = \text{wavelength} \quad m = \text{mass}$$

$$p = \text{momentum}$$

$$\text{Speed of light, } c = 3.0 \times 10^8 \text{ m s}^{-1}$$

$$\text{Planck's constant, } h = 6.63 \times 10^{-34} \text{ J s}$$

$$\text{Boltzmann's constant, } k = 1.38 \times 10^{-23} \text{ J K}^{-1}$$

$$\text{Avogadro's number} = 6.022 \times 10^{23} \text{ mol}^{-1}$$

$$\text{Electron charge, } e = -1.602 \times 10^{-19} \text{ coulomb}$$

$$1 \text{ electron volt per atom} = 96.5 \text{ kJ mol}^{-1}$$

Equilibrium Constants

$$K_a \text{ (weak acid)}$$

$$K_b \text{ (weak base)}$$

$$K_w \text{ (water)}$$

$$K_p \text{ (gas pressure)}$$

$$K_c \text{ (molar concentrations)}$$

$$S^\circ = \text{standard entropy}$$

$$H^\circ = \text{standard enthalpy}$$

$$G^\circ = \text{standard free energy}$$

$$E^\circ = \text{standard reduction potential}$$

$$T = \text{temperature}$$

$$n = \text{moles}$$

$$m = \text{mass}$$

$$q = \text{heat}$$

$$c = \text{specific heat capacity}$$

$$C_p = \text{molar heat capacity at constant pressure}$$

$$E_a = \text{activation energy}$$

$$k = \text{rate constant}$$

$$A = \text{frequency factor}$$

$$\text{Faraday's constant, } \mathcal{F} = 96,500 \text{ coulombs per mole of electrons}$$

$$\text{Gas constant, } R = 8.31 \text{ J mol}^{-1} \text{ K}^{-1}$$

$$= 0.0821 \text{ L atm mol}^{-1} \text{ K}^{-1}$$

$$= 62.4 \text{ L torr mol}^{-1} \text{ K}^{-1}$$

$$= 8.31 \text{ volt coulomb mol}^{-1} \text{ K}^{-1}$$

GO ON TO THE NEXT PAGE.

GASES, LIQUIDS, AND SOLUTIONS

$$PV = nRT$$

$$\left(P + \frac{n^2 a}{V^2}\right)(V - nb) = nRT$$

$$P_A = P_{\text{total}} \times X_A, \text{ where } X_A = \frac{\text{moles A}}{\text{total moles}}$$

$$P_{\text{total}} = P_A + P_B + P_C + \dots$$

$$n = \frac{m}{M}$$

$$K = ^\circ\text{C} + 273$$

$$\frac{P_1 V_1}{T_1} = \frac{P_2 V_2}{T_2}$$

$$D = \frac{m}{V}$$

$$u_{\text{rms}} = \sqrt{\frac{3kT}{m}} = \sqrt{\frac{3RT}{M}}$$

$$KE \text{ per molecule} = \frac{1}{2}mv^2$$

$$KE \text{ per mole} = \frac{3}{2}RT$$

$$\frac{r_1}{r_2} = \sqrt{\frac{M_2}{M_1}}$$

molarity, M = moles solute per liter solution

molality = moles solute per kilogram solvent

$$\Delta T_f = iK_f \times \text{molality}$$

$$\Delta T_b = iK_b \times \text{molality}$$

$$\pi = iMRT$$

$$A = abc$$

P = pressure

V = volume

T = temperature

n = number of moles

D = density

m = mass

v = velocity

u_{rms} = root-mean-square speed

KE = kinetic energy

r = rate of effusion

M = molar mass

π = osmotic pressure

i = van't Hoff factor

K_f = molal freezing-point depression constant

K_b = molal boiling-point elevation constant

A = absorbance

a = molar absorptivity

b = path length

c = concentration

Q = reaction quotient

I = current (amperes)

q = charge (coulombs)

t = time (seconds)

E° = standard reduction potential

K = equilibrium constant

OXIDATION-REDUCTION; ELECTROCHEMISTRY

$$Q = \frac{[C]^c [D]^d}{[A]^a [B]^b}, \text{ where } aA + bB \rightarrow cC + dD$$

$$I = \frac{q}{t}$$

$$E_{\text{cell}} = E_{\text{cell}}^\circ - \frac{RT}{n\mathcal{F}} \ln Q = E_{\text{cell}}^\circ - \frac{0.0592}{n} \log Q \text{ @ } 25^\circ\text{C}$$

$$\log K = \frac{nE^\circ}{0.0592}$$

$$\begin{aligned} \text{Gas constant, } R &= 8.31 \text{ J mol}^{-1} \text{ K}^{-1} \\ &= 0.0821 \text{ L atm mol}^{-1} \text{ K}^{-1} \\ &= 62.4 \text{ L torr mol}^{-1} \text{ K}^{-1} \\ &= 8.31 \text{ volt coulomb mol}^{-1} \text{ K}^{-1} \end{aligned}$$

$$\text{Boltzmann's constant, } k = 1.38 \times 10^{-23} \text{ J K}^{-1}$$

$$K_f \text{ for H}_2\text{O} = 1.86 \text{ K kg mol}^{-1}$$

$$K_b \text{ for H}_2\text{O} = 0.512 \text{ K kg mol}^{-1}$$

$$1 \text{ atm} = 760 \text{ mm Hg}$$

$$= 760 \text{ torr}$$

$$\text{STP} = 0.00^\circ\text{C and } 1.0 \text{ atm}$$

$$\text{Faraday's constant, } \mathcal{F} = 96,500 \text{ coulombs per mole of electrons}$$

CHEMISTRY

Section I

Time—1 hour and 30 minutes

NO CALCULATOR MAY BE USED WITH SECTION I.

Note: For all questions, assume that the temperature is 298 K, the pressure is 1.00 atmosphere, and solutions are aqueous unless otherwise specified.

Throughout the test the following symbols have the definitions specified unless otherwise noted.

T = temperature	L, mL = liter(s), milliliter(s)
P = pressure	g = gram(s)
V = volume	nm = nanometer(s)
S = entropy	atm = atmosphere(s)
H = enthalpy	mm Hg = millimeters of mercury
G = Gibbs free energy	J, kJ = joule(s), kilojoule(s)
R = molar gas constant	V = volt(s)
n = number of moles	mol = mole(s)
M = molar	
m = molal	

Part A

Directions: Each set of lettered choices below refers to the numbered statements immediately following it. Select the one lettered choice that best fits each statement and then fill in the corresponding circle on the answer sheet. A choice may be used once, more than once, or not at all in each set.

Questions 1-3 refer to the following gaseous molecules.

- (A) BeCl_2
- (B) SO_2
- (C) N_2
- (D) O_2
- (E) F_2

1. Is a polar molecule
2. Is best represented by two or more resonance forms
3. Is the molecule in which the intramolecular forces are strongest

Questions 4-5

- (A) Alpha-particle emission
- (B) Beta-particle emission
- (C) Electron capture
- (D) Gamma-ray emission
- (E) Nuclear fission

4. Is the major process by which nuclei lose excess energy without a change in atomic number
5. Accounts for the transformation of $^{207}_{81}\text{Tl}$ into $^{207}_{82}\text{Pb}$

Questions 6-8 refer to the following types of chemical or physical changes.

- (A) Oxidation-reduction reaction
- (B) Brønsted-Lowry acid-base reaction
- (C) Sublimation
- (D) Dehydration
- (E) Precipitation

6. Occurs when aqueous solutions of ammonia and vinegar are mixed
7. Occurs when $\text{Al}(s)$ and $\text{CuCl}_2(aq)$ are mixed
8. Occurs when solid sodium acetate, $\text{NaC}_2\text{H}_3\text{O}_2(s)$, is added to water

Questions 9-10 refer to the following gas molecules at the conditions indicated.

- (A) $\text{H}_2(g)$ molecules at 10^{-3} atm and 200°C
- (B) $\text{O}_2(g)$ molecules at 20 atm and 200°C
- (C) $\text{SO}_2(g)$ molecules at 20 atm and 200°C
- (D) $\text{NH}_3(g)$ molecules at 20 atm and 200°C
- (E) $\text{NH}_3(g)$ molecules at 20 atm and 300°C

- 9. Behave most like an ideal gas
- 10. Have lowest root-mean-square speed

Questions 11-13

- (A) Cs
- (B) Ag
- (C) Pb
- (D) Br
- (E) Se

- 11. Has the highest electronegativity
- 12. Has the lowest first-ionization energy
- 13. Has the largest atomic radius

Questions 14-16 refer to the following compounds.

- (A) $\text{CH}_3\text{CH}_2\text{CH}_2\text{CH}_3$
- (B) $\text{CH}_3\text{CH}_2\text{CH}_2\text{OH}$
- (C) CH_3COCH_3
- (D) CH_3COOH
- (E) $\text{CH}_3\text{CH}_2\text{CH}_2\text{NH}_2$

14. Is isomeric with $\text{CH}_3\text{CH}_2\text{CHO}$

15. Dissolves in water to form an acidic solution

16. Is the LEAST soluble in water

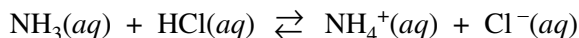
Part B

Directions: Each of the questions or incomplete statements below is followed by five suggested answers or completions. Select the one that is best in each case and then fill in the corresponding circle on the answer sheet.

17. A certain crystalline substance that has a low melting point does not conduct electricity in solution or when melted. This substance is likely to be
- (A) a covalent network solid
 - (B) a metallic solid
 - (C) a polymer
 - (D) an ionic solid
 - (E) a molecular solid
18. Solid $\text{Al}(\text{NO}_3)_3$ is added to distilled water to produce a solution in which the concentration of nitrate, $[\text{NO}_3^-]$, is 0.10 M . What is the concentration of aluminum ion, $[\text{Al}^{3+}]$, in this solution?
- (A) 0.010 M
 - (B) 0.033 M
 - (C) 0.066 M
 - (D) 0.10 M
 - (E) 0.30 M
19. Which of the following is a weak acid in aqueous solution?
- (A) HCl
 - (B) HClO_4
 - (C) HNO_3
 - (D) H_2S
 - (E) H_2SO_4
20. In 1.00 mol of potassium zirconium sulfate trihydrate, $\text{K}_4\text{Zr}(\text{SO}_4)_4 \cdot 3\text{H}_2\text{O}$, there are
- (A) $3 \times 6.02 \times 10^{23}$ hydrogen atoms
 - (B) 6.02×10^{23} sulfur atoms
 - (C) $4 \times 6.02 \times 10^{23}$ potassium atoms
 - (D) 4 moles of oxygen atoms
 - (E) 4 moles of zirconium atoms
- $\text{X} + 2\text{Y} \rightarrow \text{Z} + 3\text{Q}$
21. For the reaction represented above, the initial rate of decrease in $[\text{X}]$ was $2.8 \times 10^{-3}\text{ mol L}^{-1}\text{ s}^{-1}$. What was the initial rate of decrease in $[\text{Y}]$?
- (A) $7.0 \times 10^{-4}\text{ mol L}^{-1}\text{ s}^{-1}$
 - (B) $1.4 \times 10^{-3}\text{ mol L}^{-1}\text{ s}^{-1}$
 - (C) $2.8 \times 10^{-3}\text{ mol L}^{-1}\text{ s}^{-1}$
 - (D) $5.6 \times 10^{-3}\text{ mol L}^{-1}\text{ s}^{-1}$
 - (E) $1.1 \times 10^{-2}\text{ mol L}^{-1}\text{ s}^{-1}$
22. To determine the percentage of water in a hydrated salt, a student heated a 1.2346 g sample of the salt for 30 minutes; when cooled to room temperature, the sample weighed 1.1857 g . After the sample was heated for an additional 10 minutes and again cooled to room temperature, the sample weighed 1.1632 g . Which of the following should the student do next?
- (A) Use the smallest mass value to calculate the percentage of water in the hydrated salt.
 - (B) Repeat the experiment with a new sample of the same mass and average the results.
 - (C) Repeat the experiment with a new sample that has a different mass.
 - (D) Reheat the sample until its mass is constant.
 - (E) Use the average of the mass values obtained after the two heatings to calculate the percentage of water in the hydrated salt.

23. Which of the following statements about atoms is NOT correct?

- (A) Atoms are electrically neutral because they have the same number of protons and electrons.
- (B) All atoms of a given element must have the same number of protons, neutrons, and electrons.
- (C) Most of the volume of an atom contains only electrons.
- (D) The nucleus is positively charged.
- (E) Almost all of the mass of an atom is in the nucleus.

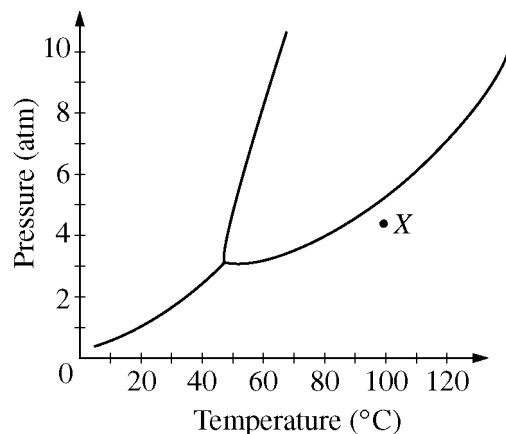


24. The Brønsted-Lowry bases in the reaction represented above are

- (A) $\text{NH}_3(aq)$ and $\text{NH}_4^+(aq)$
- (B) $\text{NH}_3(aq)$ and $\text{Cl}^-(aq)$
- (C) $\text{NH}_3(aq)$ and $\text{HCl}(aq)$
- (D) $\text{HCl}(aq)$ and $\text{NH}_4^+(aq)$
- (E) $\text{HCl}(aq)$ and $\text{Cl}^-(aq)$

25. When 6.0 L of $\text{He}(g)$ and 10. L of $\text{N}_2(g)$, both at 0°C and 1.0 atm, are pumped into an evacuated 4.0 L rigid container, the final pressure in the container at 0°C is

- (A) 2.0 atm
- (B) 4.0 atm
- (C) 6.4 atm
- (D) 8.8 atm
- (E) 16 atm



26. Shown above is the phase diagram of a pure substance. The substance under the conditions corresponding to point X on the diagram is cooled to 40°C while the pressure remains constant. As the substance cools, the phase of the substance changes from

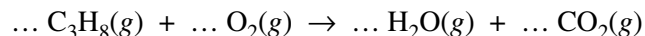
- (A) gas to liquid to solid
- (B) gas to solid to liquid
- (C) solid to liquid to gas
- (D) liquid to solid to gas
- (E) liquid to gas to solid

27. Oxygen is acting as an oxidizing agent in all of the following reactions EXCEPT

- (A) $2\text{C}(s) + \text{O}_2(g) \rightarrow 2\text{CO}(g)$
- (B) $\text{S}(s) + \text{O}_2(g) \rightarrow \text{SO}_2(g)$
- (C) $2\text{F}_2(g) + \text{O}_2(g) \rightarrow 2\text{OF}_2(g)$
- (D) $2\text{Na}(s) + \text{O}_2(g) \rightarrow \text{Na}_2\text{O}_2(s)$
- (E) $2\text{Mg}(s) + \text{O}_2(g) \rightarrow 2\text{MgO}(s)$

28. What is the maximum number of moles of Al_2O_3 that can be produced by the reaction of 0.40 mol of Al with 0.40 mol of O_2 ?

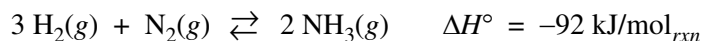
- (A) 0.10 mol
- (B) 0.20 mol
- (C) 0.27 mol
- (D) 0.33 mol
- (E) 0.40 mol



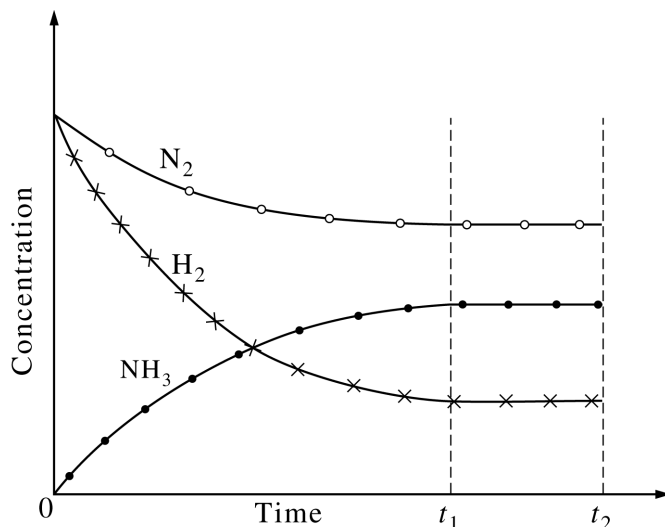
29. When the equation for the reaction represented above is balanced and all coefficients are reduced to the lowest whole-number terms, the coefficient for $\text{O}_2(g)$ is
- (A) 1
(B) 2
(C) 3
(D) 5
(E) 6
30. A 0.1 *M* solution of which of the following is colorless?
- (A) MgCl_2
(B) $\text{Ni}(\text{NO}_3)_2$
(C) Na_2CrO_4
(D) KMnO_4
(E) CuSO_4
31. Under which of the following conditions can an endothermic reaction be thermodynamically favorable?
- (A) ΔG is positive
(B) ΔS is negative
(C) $T\Delta S > \Delta H$
(D) $T\Delta S = 0$
(E) There are no conditions under which an endothermic reaction can be thermodynamically favorable.
32. The vapor pressure of pure water at 25°C is 24.0 mm Hg. What is the expected vapor pressure at 25°C of an ideal solution of a nonvolatile nonelectrolyte in which the mole fraction of water is 0.900 ?
- (A) 1.48 mm Hg
(B) 2.40 mm Hg
(C) 21.6 mm Hg
(D) 24.0 mm Hg
(E) 26.7 mm Hg
33. Which of the following salts is LEAST soluble in water?
- (A) NiS
(B) MgCl_2
(C) K_2CrO_4
(D) $\text{Al}_2(\text{SO}_4)_3$
(E) $\text{Pb}(\text{NO}_3)_2$
34. Which of the following is the best piece of laboratory glassware for preparing 500.0 mL of an aqueous solution of a solid?
- (A) Volumetric flask
(B) Erlenmeyer flask
(C) Test tube
(D) Graduated beaker
(E) Graduated cylinder

Questions 35-36 refer to the experiment described below.

H_2 gas and N_2 gas were placed in a rigid vessel and allowed to reach equilibrium in the presence of a catalyst according to the following equation.



The diagram below shows how the concentrations of H_2 , N_2 , and NH_3 in this system changed over time.



35. Which of the following was true for the system between time t_1 and time t_2 ?

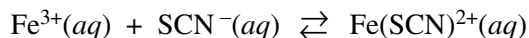
- (A) The concentration of N_2 decreased.
- (B) The temperature of the system decreased.
- (C) The number of effective collisions between H_2 and N_2 was zero.
- (D) The rates of the forward and reverse reactions were equal.
- (E) The rate of formation of NH_3 molecules was equal to the rate of disappearance of H_2 molecules.

36. More NH_3 gas is added to the system at time t_2 while the temperature is held constant. Which of the following will most likely occur?

- (A) The value of the equilibrium constant will increase.
- (B) The value of the equilibrium constant will decrease.
- (C) The total pressure in the container will decrease.
- (D) The amount of N_2 will increase.
- (E) The amount of H_2 will decrease.

37. When heated, metallic carbonates generally produce

- (A) metallic peroxide + CO
- (B) metal + CO + O₂
- (C) metallic hydroxide + CO₂
- (D) metallic oxalate + O₂
- (E) metallic oxide + CO₂



38. For the reaction represented above, the value of the equilibrium constant, K_{eq} , is 240 at 25°C. From this information, correct deductions about the reaction at 25°C include which of the following?

- I. The reaction is quite rapid.
- II. The product is favored over the reactants at equilibrium.
- III. The reaction is endothermic.

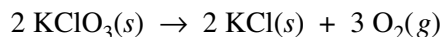
- (A) I only
- (B) II only
- (C) I and II only
- (D) II and III only
- (E) I, II, and III

39. The volume of water that must be added in order to dilute 40 mL of 9.0 M HCl to a concentration of 6.0 M is closest to

- (A) 10 mL
- (B) 20 mL
- (C) 30 mL
- (D) 40 mL
- (E) 60 mL

40. Which of the following statements best explains why an increase in temperature of 5-10 Celsius degrees can substantially increase the rate of a chemical reaction?

- (A) The activation energy for the reaction is lowered.
- (B) The number of effective collisions between reactant particles is increased.
- (C) The rate of the reverse reaction is increased.
- (D) ΔH for the reaction is lowered.
- (E) ΔG for the reaction becomes more positive.



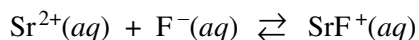
41. What is the percentage yield of O₂ if 12.3 g of KClO₃ (molar mass 123 g) is decomposed to produce 3.2 g of O₂ (molar mass 32 g) according to the equation above?

- (A) 100%
- (B) 67%
- (C) 50%
- (D) 33%
- (E) 10%

42. When a strong acid is titrated with a strong base using phenolphthalein as an indicator, the color changes abruptly at the endpoint of the titration and can be switched back and forth by the addition of only one drop of acid or base. The reason for the abruptness of this color change is that

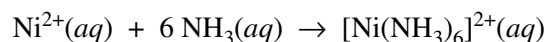
- (A) a large change in pH occurs near the endpoint of the titration
- (B) a buffer solution exists at the endpoint of the titration
- (C) phenolphthalein is a strong proton donor
- (D) the pH of water is very resistant to change
- (E) phenolphthalein is much more sensitive to the pH of a solution than most other indicators

43. A 1 mol sample of zinc can reduce the greatest number of moles of which of the following ions?
- (A) Al^{3+}
 (B) Pb^{2+}
 (C) Ag^+
 (D) Cl^-
 (E) N^{3-}
44. At 298 K and 1 atm, bromine is a liquid with a high vapor pressure, whereas chlorine is a gas. This provides evidence that, under these conditions, the
- (A) forces among Br_2 molecules are greater than those among Cl_2 molecules
 (B) forces among Br_2 molecules are weaker than the Br–Br bond
 (C) forces among Cl_2 molecules are stronger than the Cl–Cl bond
 (D) Br–Br bond is stronger than the Cl–Cl bond
 (E) Br–Br bond is weaker than the Cl–Cl bond
45. The value of K_{sp} for PbCl_2 is 1.6×10^{-5} . What is the lowest concentration of $\text{Cl}^-(aq)$ that would be needed to begin precipitation of $\text{PbCl}_2(s)$ in $0.010\text{ M Pb}(\text{NO}_3)_2$?
- (A) $1.6 \times 10^{-7}\text{ M}$
 (B) $4.0 \times 10^{-4}\text{ M}$
 (C) $1.6 \times 10^{-3}\text{ M}$
 (D) $2.6 \times 10^{-3}\text{ M}$
 (E) $4.0 \times 10^{-2}\text{ M}$
46. Which of the following aqueous solutions has the lowest freezing point?
- (A) 0.2 m NaCl
 (B) 0.2 m CaCl_2
 (C) $0.2\text{ m H}_2\text{SO}_4$
 (D) 0.2 m NH_3
 (E) $0.2\text{ m Al}(\text{NO}_3)_3$
- Step 1: $\text{NO}(g) + \text{O}_3(g) \rightarrow \text{NO}_2(g) + \text{O}_2(g)$
 Step 2: $\text{NO}_2(g) + \text{O}(g) \rightarrow \text{NO}(g) + \text{O}_2(g)$
47. A reaction mechanism for the destruction of ozone, $\text{O}_3(g)$, is represented above. In the overall reaction, $\text{NO}(g)$ is best described as
- (A) an inhibitor
 (B) a catalyst
 (C) a reactant
 (D) an intermediate
 (E) a product
48. When a buret is rinsed before a titration, which of the techniques below is the best procedure?
- (A) Rinse the buret one time with some of the titrant solution.
 (B) Rinse the buret one time with some of the titrant solution and then dry the buret in an oven.
 (C) Rinse the buret two times: once with some of the titrant solution, then once with distilled water.
 (D) Rinse the buret two times: each time with some of the titrant solution.
 (E) Rinse the buret two times: each time with distilled water.



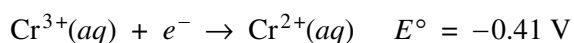
49. At 25°C, the equilibrium constant for the reaction represented above has a value of 1.3. At 50°C, the value of the equilibrium constant is less than 1.3. Based on this information, which of the following must be correct?

- (A) The reaction rate decreases as the temperature is increased.
- (B) The reaction is thermodynamically favorable only at temperatures above 25°C.
- (C) At 25°C, ΔG° for the reaction is positive.
- (D) At 25°C, ΔS° for the reaction is positive.
- (E) At 25°C, ΔH° for the reaction is negative.



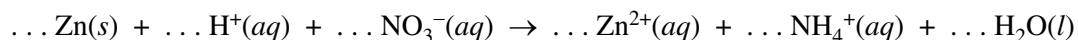
50. The reaction represented above is best classified as

- (A) a Lewis acid-base reaction
- (B) a Brønsted-Lowry acid-base reaction
- (C) an Arrhenius acid-base reaction
- (D) an oxidation-reduction reaction
- (E) a precipitation reaction



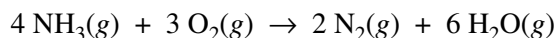
51. According to the half-reactions represented above, which of the following occurs in aqueous solutions under standard conditions?

- (A) $\text{Cu}^{2+}(\text{aq}) + \text{Cr}^{3+}(\text{aq}) \rightarrow \text{Cu}(\text{s}) + \text{Cr}^{2+}(\text{aq})$
- (B) $\text{Cu}^{2+}(\text{aq}) + 2 \text{Cr}^{2+}(\text{aq}) \rightarrow \text{Cu}(\text{s}) + 2 \text{Cr}^{3+}(\text{aq})$
- (C) $\text{Cu}(\text{s}) + 2 \text{Cr}^{3+}(\text{aq}) \rightarrow \text{Cu}^{2+}(\text{aq}) + 2 \text{Cr}^{2+}(\text{aq})$
- (D) $\text{Cu}(\text{s}) + \text{Cr}^{3+}(\text{aq}) \rightarrow \text{Cu}^{2+}(\text{aq}) + \text{Cr}^{2+}(\text{aq})$
- (E) $2 \text{Cu}^{2+}(\text{aq}) + \text{Cr}^{3+}(\text{aq}) \rightarrow 2 \text{Cu}(\text{s}) + \text{Cr}^{2+}(\text{aq})$



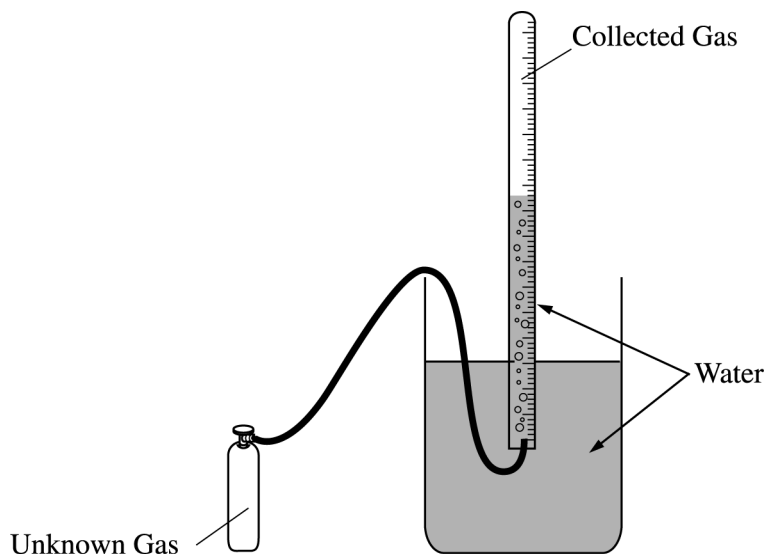
52. When the equation above is balanced and all coefficients are reduced to lowest whole number terms, the coefficient for $\text{Zn}(\text{s})$ is

- (A) 2
- (B) 4
- (C) 6
- (D) 10
- (E) 14



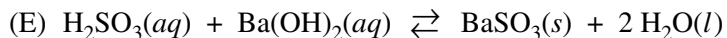
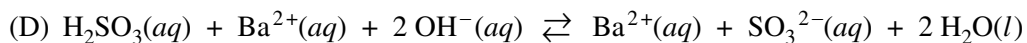
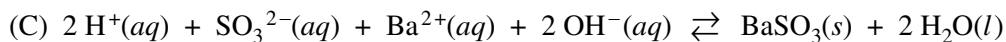
53. If the standard molar heats of formation of ammonia, $\text{NH}_3(g)$, and gaseous water, $\text{H}_2\text{O}(g)$, are -46 kJ/mol and -242 kJ/mol , respectively, what is the value of ΔH_{298}° for the reaction represented above?
- (A) $-190 \text{ kJ/mol}_{\text{rxn}}$
 (B) $-290 \text{ kJ/mol}_{\text{rxn}}$
 (C) $-580 \text{ kJ/mol}_{\text{rxn}}$
 (D) $-1,270 \text{ kJ/mol}_{\text{rxn}}$
 (E) $-1,640 \text{ kJ/mol}_{\text{rxn}}$
54. When a magnesium wire is dipped into a solution of lead(II) nitrate, a black deposit forms on the wire. Which of the following can be concluded from this observation?
- (A) The standard reduction potential, E° , for $\text{Pb}^{2+}(aq)$ is greater than that for $\text{Mg}^{2+}(aq)$.
 (B) $\text{Mg}(s)$ is less easily oxidized than $\text{Pb}(s)$.
 (C) An external source of potential must have been supplied.
 (D) The magnesium wire will be the cathode of a Mg/Pb cell.
 (E) $\text{Pb}(s)$ can spontaneously displace $\text{Mg}^{2+}(aq)$ from solution.
55. Which of the molecules represented below contains carbon with sp^2 hybridization?
- (A) CH_4
 (B) CH_2Cl_2
 (C) C_2H_6
 (D) $\text{C}_2\text{H}_2\text{Cl}_2$
 (E) $\text{C}_2\text{H}_4\text{Cl}_2$
56. A chemical supply company sells a concentrated solution of aqueous H_2SO_4 (molar mass 98 g mol^{-1}) that is 50. percent H_2SO_4 by mass. At 25°C , the density of the solution is 1.4 g mL^{-1} . What is the molarity of the H_2SO_4 solution at 25°C ?
- (A) 1.8 M
 (B) 3.6 M
 (C) 5.1 M
 (D) 7.1 M
 (E) 14 M
57. A reaction produces a colorless gas, which is collected by water displacement. A glowing splint inserted into a bottle full of the gas is extinguished. The gas could be
- (A) N_2
 (B) NO_2
 (C) O_2
 (D) Br_2
 (E) Cl_2
58. A solution of methanol, CH_3OH , in water is prepared by mixing together 128 g of methanol and 108 g of water. The mole fraction of methanol in the solution is closest to
- (A) 0.80
 (B) 0.60
 (C) 0.50
 (D) 0.40
 (E) 0.20
59. A sample of a compound contains 3.21 g of sulfur and 11.4 g of fluorine. Which of the following represents the empirical formula of the compound?
- (A) SF_2
 (B) SF_3
 (C) SF_4
 (D) SF_5
 (E) SF_6

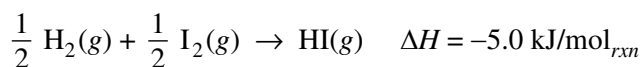
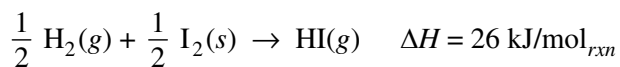
60. Of the following, the best explanation for the fact that most gases are easily compressed is that the molecules in a gas
- (A) are in constant motion
 - (B) are relatively far apart
 - (C) have relatively small masses
 - (D) have a real, nonzero volume
 - (E) move slower as temperature decreases
61. Given that the density of $\text{Hg}(l)$ at 0°C is about 14 g mL^{-1} , which of the following is closest to the volume of one mole of $\text{Hg}(l)$ at this temperature?
- (A) 0.070 mL
 - (B) 0.14 mL
 - (C) 1.4 mL
 - (D) 14 mL
 - (E) 28 mL



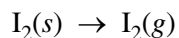
62. A sample of an unknown gas from a cylinder is collected over water in the apparatus shown above. After all the gas sample has been collected, the water levels inside and outside the gas collection tube are made the same. Measurements that must be made to calculate the molar mass of the gas include all of the following EXCEPT
- (A) atmospheric pressure
 - (B) temperature of the water
 - (C) volume of gas in the gas-collection tube
 - (D) initial and final mass of the gas cylinder
 - (E) mass of the water in the apparatus

63. Addition of sulfurous acid (a weak acid) to barium hydroxide (a strong base) results in the formation of a precipitate. The net ionic equation for this reaction is



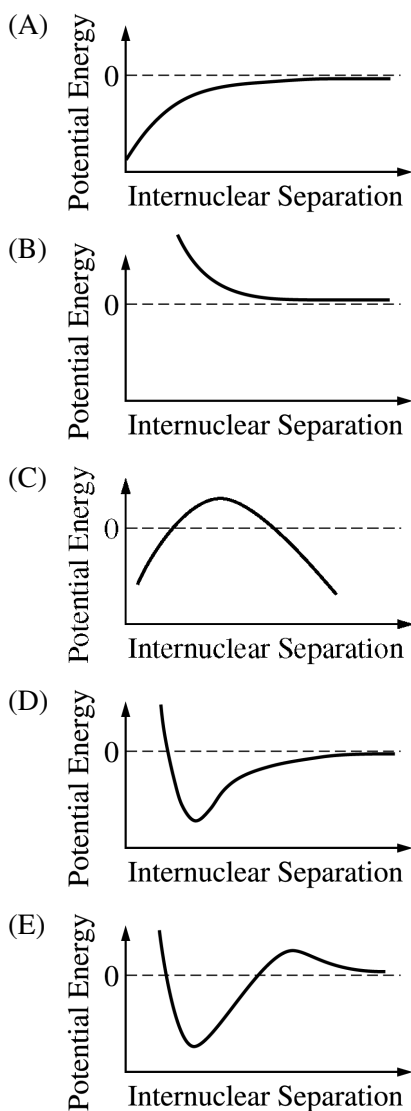


64. Based on the information above, what is the enthalpy change for the sublimation of iodine, represented below?



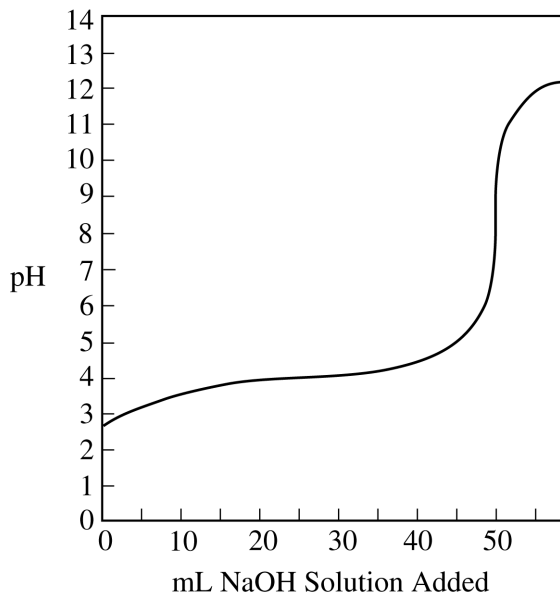
- (A) 15 kJ/mol_{rxn}
 (B) 21 kJ/mol_{rxn}
 (C) 31 kJ/mol_{rxn}
 (D) 42 kJ/mol_{rxn}
 (E) 62 kJ/mol_{rxn}

65. Which of the following graphs correctly shows the relationship between potential energy and internuclear separation for two hydrogen atoms?



66. Which of the following compounds is LEAST likely to exist?

- (A) PCl_5
- (B) PBr_3
- (C) NF_3
- (D) NI_5
- (E) SbF_5



67. The graph above shows the titration curve that resulted when a sample of 0.1 M monoprotic acid was titrated with a solution of NaOH . Based on the graph, the $\text{p}K_a$ of the acid is closest to

- (A) 3.0
- (B) 4.0
- (C) 6.0
- (D) 8.0
- (E) 12.0

68. The rate law for the reaction of nitrogen dioxide and chlorine is found to be $\text{rate} = k [\text{NO}_2]^2 [\text{Cl}_2]$.

By what factor does the rate of the reaction change when the concentrations of both NO_2 and Cl_2 are doubled?

- (A) 2
- (B) 3
- (C) 4
- (D) 6
- (E) 8

69. When a student prepares an aqueous solution containing the five cations $\text{Ag}^+(aq)$, $\text{Hg}_2^{2+}(aq)$, $\text{Cu}^{2+}(aq)$, $\text{Mn}^{2+}(aq)$, and $\text{Ba}^{2+}(aq)$, the student observes that no precipitates form in the solution. Which of the following could be the identity of the anion in the solution?

- (A) $\text{Cl}^-(aq)$
- (B) $\text{CO}_3^{2-}(aq)$
- (C) $\text{CrO}_4^{2-}(aq)$
- (D) $\text{NO}_3^-(aq)$
- (E) $\text{SO}_4^{2-}(aq)$

70. What is the molarity of $\text{I}^-(aq)$ in a solution that contains 34 g of SrI_2 (molar mass 341 g) in 1.0 L of the solution?

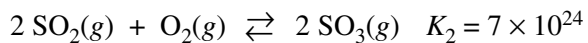
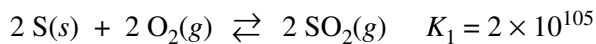
- (A) 0.034 M
- (B) 0.068 M
- (C) 0.10 M
- (D) 0.20 M
- (E) 0.68 M

71. Which of the following compounds contains both ionic and covalent bonds?

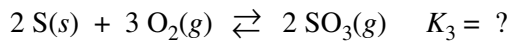
- (A) SO_3
- (B) $\text{C}_2\text{H}_5\text{OH}$
- (C) MgF_2
- (D) H_2S
- (E) NH_4Cl

72. Some pollutant gases in the atmosphere act as contributors to the formation of acid rain, a serious environmental problem. An example of such a gas is

- (A) N_2
- (B) O_2
- (C) H_2O
- (D) NO_2
- (E) CH_4



73. Given the value of the equilibrium constants K_1 and K_2 for the reactions represented above, what is the value of the equilibrium constant, K_3 , for the following reaction?



- (A) 1×10^{130}
- (B) 3×10^{80}
- (C) 1×10^{65}
- (D) 2×10^{40}
- (E) 7×10^{24}

74. Which of the following molecules is nonpolar but has polar covalent bonds?

- (A) N_2
- (B) H_2O_2
- (C) H_2O
- (D) CCl_4
- (E) CH_2Cl_2

75. A 0.10 M solution of which of the following salts is most basic?

- (A) LiNO_3
- (B) Na_2SO_4
- (C) CaCl_2
- (D) $\text{Al}(\text{NO}_3)_3$
- (E) K_2CO_3

END OF SECTION I

**IF YOU FINISH BEFORE TIME IS CALLED,
YOU MAY CHECK YOUR WORK ON THIS SECTION.**

DO NOT GO ON TO SECTION II UNTIL YOU ARE TOLD TO DO SO.

MAKE SURE YOU HAVE DONE THE FOLLOWING.

- **PLACED YOUR AP NUMBER LABEL ON YOUR ANSWER SHEET**
- **WRITTEN AND GRIDDED YOUR AP NUMBER CORRECTLY ON YOUR ANSWER SHEET**
- **TAKEN THE AP EXAM LABEL FROM THE FRONT OF THIS BOOKLET AND PLACED IT ON YOUR ANSWER SHEET**

CHEMISTRY
Section II
(Total time—95 minutes)

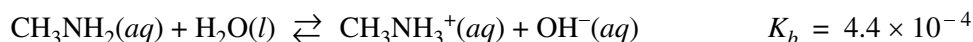
Part A

Time—55 minutes

YOU MAY USE YOUR CALCULATOR FOR PART A.

CLEARLY SHOW THE METHOD USED AND THE STEPS INVOLVED IN ARRIVING AT YOUR ANSWERS. It is to your advantage to do this, since you may obtain partial credit if you do and you will receive little or no credit if you do not. Attention should be paid to significant figures. Be sure to write all your answers to the questions on the lined pages following each question in this booklet.

Answer Questions 1, 2, and 3. The Section II score weighting for each question is 20 percent.



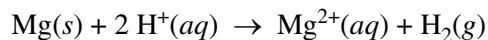
1. Methylamine, CH_3NH_2 , is a weak base that reacts with water according to the equation above. A student obtains a 50.0 mL sample of a methylamine solution and determines the pH of the solution to be 11.77.
- (a) Write the expression for the equilibrium constant, K_b , for methylamine.
 - (b) Calculate the molar concentration of OH^- in the 50.0 mL sample of the methylamine solution.
 - (c) Calculate the initial molar concentration of $\text{CH}_3\text{NH}_2(aq)$ in the solution before it reacted with water and equilibrium was established.

The 50.0 mL sample of the methylamine solution is titrated with an HCl solution of unknown concentration. The equivalence point of the titration is reached after a volume of 36.0 mL of the HCl solution is added. The pH of the solution at the equivalence point is 5.98.

- (d) Write the net-ionic equation that represents the reaction that takes place during the titration.
- (e) Calculate the concentration of the HCl solution used to titrate the methylamine.
- (f) Using the axes provided, sketch the titration curve that results from the titration described above. On the graph, clearly label the equivalence point of the titration.



[illegible]



2. A student performs an experiment to determine the volume of hydrogen gas produced when a given mass of magnesium reacts with excess $\text{HCl}(aq)$, as represented by the net ionic equation above. The student begins with a 0.0360 g sample of pure magnesium and a solution of 2.0 M $\text{HCl}(aq)$.

- (a) Calculate the number of moles of magnesium in the 0.0360 g sample.
- (b) Calculate the number of moles of $\text{HCl}(aq)$ needed to react completely with the sample of magnesium.

As the magnesium reacts, the hydrogen gas produced is collected by water displacement at 23.0°C. The pressure of the gas in the collection tube is measured to be 749 torr.

- (c) Given that the equilibrium vapor pressure of water is 21 torr at 23.0°C, calculate the pressure that the $\text{H}_2(g)$ produced in the reaction would have if it were dry.
- (d) Calculate the volume, in liters measured at the conditions in the laboratory, that the $\text{H}_2(g)$ produced in the reaction would have if it were dry.
- (e) The laboratory procedure specified that the concentration of the HCl solution be 2.0 M , but only 12.3 M HCl solution was available. Describe the steps for safely preparing 50.0 mL of 2.0 M $\text{HCl}(aq)$ using 12.3 M HCl solution and materials selected from the list below. Show any necessary calculation(s).

10.0 mL graduated cylinder

Distilled water

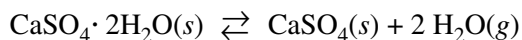
250 mL beakers

Balance

50.00 mL volumetric flask

Dropper

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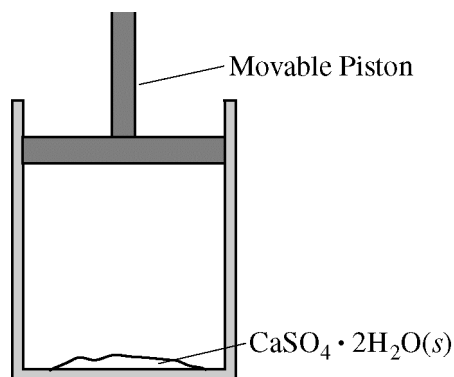
3. The hydrate $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}(s)$ can be heated to form the anhydrous salt, $\text{CaSO}_4(s)$, as shown by the reaction represented above.

(a) Using the data in the table below, calculate the value of ΔG° , in $\text{kJ/mol}_{\text{rxn}}$, for the reaction at 298 K.

Substance	ΔG_f° at 298 K (kJ/mol)
$\text{CaSO}_4 \cdot 2\text{H}_2\text{O}(s)$	-1795.70
$\text{CaSO}_4(s)$	-1320.30
$\text{H}_2\text{O}(g)$	-228.59

(b) Given that the value of ΔH° for the reaction at 298 K is $+105 \text{ kJ/mol}_{\text{rxn}}$, calculate the value of ΔS° for the reaction at 298 K. Include units with your answer.

A sample of $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}(s)$ is placed in a cylinder with a movable piston as shown in the diagram below. The air above the solid is at 1.00 atm and is initially dry (partial pressure of $\text{H}_2\text{O}(g) = 0 \text{ atm}$).



- (c) Write the expression for the equilibrium constant, K_p , for the reaction.
- (d) Given that the equilibrium constant, K_p , is 6.4×10^{-4} at 298 K, determine the partial pressure, in atm, of water vapor in the cylinder at equilibrium at 298 K.
- (e) If the volume of the system is reduced to one-half of its original volume and the system is allowed to reestablish equilibrium at 298 K, what will be the pressure, in atm, of the water vapor at the new volume? Justify your answer.

In the laboratory, the hydrate $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}(s)$ can be heated in a crucible to completely drive off the water of hydration to form the anhydrous salt, $\text{CaSO}_4(s)$.

- (f) A 2.49 g sample of pure $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}(s)$ is heated several times until the mass is constant. Calculate the mass, in grams, of the solid that remains after the dehydration reaction is complete.

[illegible]

CHEMISTRY

Part B

Time—40 minutes

NO CALCULATORS MAY BE USED FOR PART B.

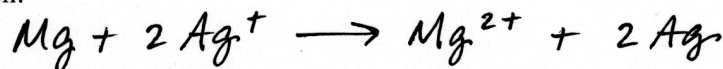
Answer Question 4 below. The Section II score weighting for this question is 10 percent.

4. For each of the following three reactions, in part (i) write a balanced equation for the reaction and in part (ii) answer the question about the reaction. In part (i), coefficients should be in terms of lowest whole numbers. Assume that solutions are aqueous unless otherwise indicated. Represent substances in solutions as ions if the substances are extensively ionized. Omit formulas for any ions or molecules that are unchanged by the reaction. You may use the empty space at the bottom of the next page for scratch work, but only equations that are written in the answer boxes provided will be scored.

EXAMPLE:

A strip of magnesium metal is added to a solution of silver(I) nitrate.

(i) Balanced equation:



(ii) Which substance is oxidized in the reaction?

Mg is oxidized.

- (a) Solutions of ethanoic (acetic) acid and lithium hydroxide are combined.

(i) Balanced equation:

- (ii) If the ethanoic acid is titrated with the lithium hydroxide, is the pH at the equivalence point equal to 7, less than 7, or greater than 7? Explain.

B B B B B B B B B B B B B

(b) Solutions of iron(III) chloride and tin(II) chloride are combined.

(i) Balanced equation:

(ii) Identify the species that is reduced in the reaction.

(c) Solutions of silver(I) nitrate and sodium phosphate are combined, forming a precipitate.

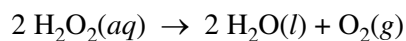
(i) Balanced equation:

(ii) If 5 mol of silver(I) nitrate and 3 mol of sodium phosphate react as completely as possible, which reactant, if any, is present in excess? Justify your answer.

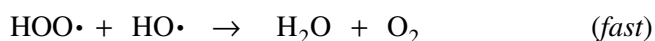
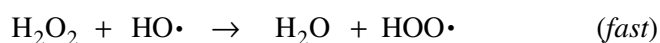
YOU MAY USE THE SPACE BELOW FOR SCRATCH WORK, BUT ONLY EQUATIONS THAT ARE WRITTEN IN THE ANSWER BOXES PROVIDED WILL BE SCORED.

Answer Question 5 and Question 6. The Section II score weighting for these questions is 15 percent each.

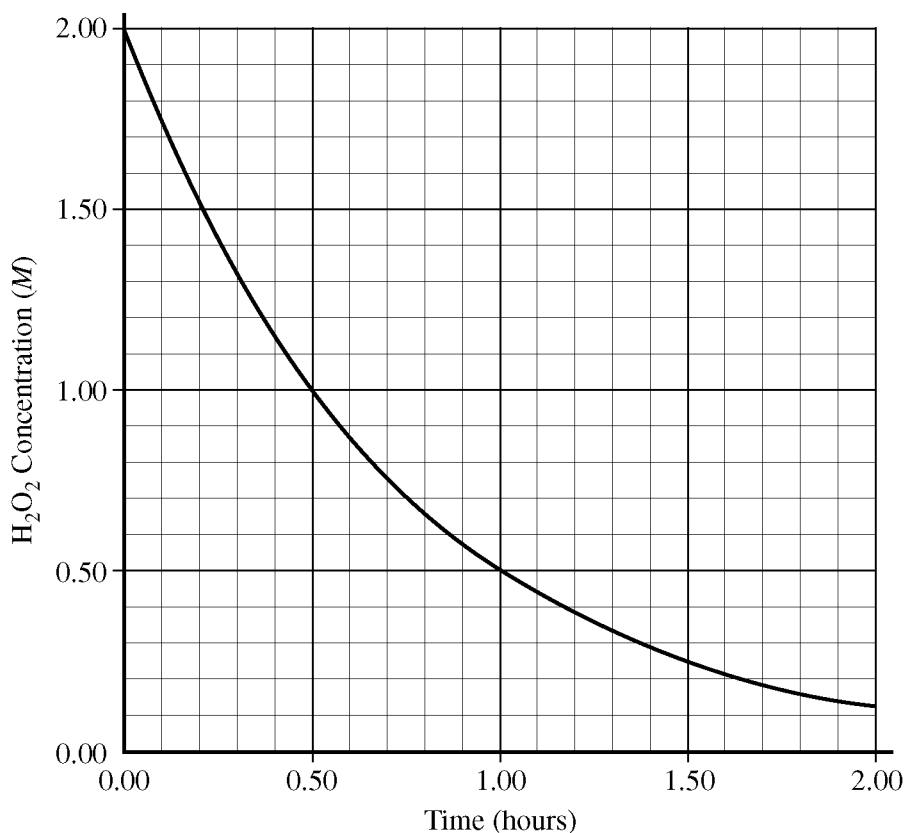
Your responses to these questions will be scored on the basis of the accuracy and relevance of the information cited. Explanations should be clear and well organized. Examples and equations may be included in your responses where appropriate. Specific answers are preferable to broad, diffuse responses.



5. The decomposition of hydrogen peroxide to form water and oxygen gas is represented by the equation above. A proposed mechanism for the reaction, which involves the free radicals $\text{HO}\cdot$ and $\text{HOO}\cdot$, is represented by the three equations below.



- (a) Write the rate law consistent with the proposed mechanism above.
- (b) The rate of the decomposition reaction was studied in an experiment, and the resulting data were plotted in the graph below.



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Using the graph, determine the time, in hours, needed for the concentration of H_2O_2 to change from

- (i) 1.50 *M* to 0.75 *M*
(ii) 0.80 *M* to 0.40 *M*

(c) The experimental data are consistent with the proposed mechanism. Explain.

An electrochemical cell based on the decomposition of H_2O_2 can be constructed based on the half-reactions in the table below.

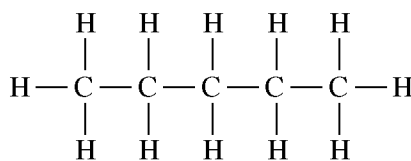
Half-Reaction	Standard Reduction Potential, E°
$\text{H}_2\text{O}_2 + 2 e^- \rightarrow 2 \text{OH}^-$	0.88 V
$\text{O}_2 + 2 \text{H}_2\text{O} + 2 e^- \rightarrow \text{H}_2\text{O}_2 + 2 \text{OH}^-$	-0.15 V

- (d) Calculate the value of the standard cell potential, E° , for the cell.
- (e) Indicate whether ΔG° for the decomposition reaction is greater than 0, less than 0, or equal to 0. Justify your answer.
- (f) The decomposition of $\text{H}_2\text{O}_2(aq)$ is slow at 298 K, but a suitable catalyst greatly increases the rate of the decomposition reaction.
- (i) Draw a circle around each of the quantities below that has a different value for the catalyzed reaction than for the uncatalyzed reaction.
- K_{eq} ΔG° ΔH° E_a
- (ii) For any quantity that you circled above, indicate whether its value is greater or less for the catalyzed reaction than for the uncatalyzed reaction. Explain why.

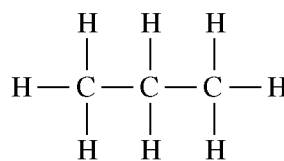
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ADDITIONAL PAGE FOR ANSWERING QUESTION 5

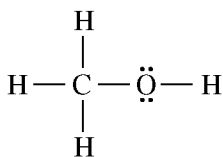
6. Answer the following questions in terms of principles of chemical bonding and intermolecular forces. In each explanation where a comparison is to be made, a complete answer must include a discussion of both substances. The following complete Lewis electron-dot diagrams may be useful in answering parts of this question.



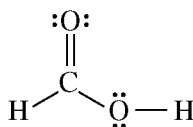
Pentane



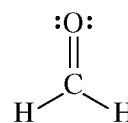
Propane



Methanol



Methanoic (formic) acid



Methanal (formaldehyde)

- (a) At 1 atm and 298 K, pentane is a liquid whereas propane is a gas. Explain.
- (b) At 1 atm and 298 K, methanol is a liquid whereas propane is a gas. Explain.
- (c) Indicate the hybridization of the carbon atom in each of the following:
- (i) Methanol
 - (ii) Methanoic (formic) acid
- (d) Draw the complete Lewis electron-dot diagram for a molecule of propanoic acid, $\text{HC}_3\text{H}_5\text{O}_2$.
- (e) Explain the following observations about the two carbon-oxygen bonds in the methanoate (formate) anion, HCO_2^- . You may draw a Lewis electron-dot diagram (or diagrams) of the methanoate ion as part of your explanations.
- (i) The two carbon-oxygen bonds in the methanoate (formate) anion, HCO_2^- , have the same length.
 - (ii) The length of the carbon-oxygen bonds in the methanoate (formate) anion, HCO_2^- , is intermediate between the length of the carbon-oxygen bond in methanol and the length of the carbon-oxygen bond in methanal.

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ADDITIONAL PAGE FOR ANSWERING QUESTION 6

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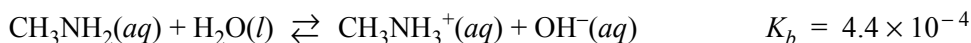
**Answer Key for AP Chemistry
Practice Exam, Section I**

Multiple-Choice Questions	
Question #	Key
1	B
2	B
3	C
4	D
5	B
6	B
7	A
8	B
9	A
10	C
11	D
12	A
13	A
14	C
15	D
16	A
17	E
18	B
19	D
20	C
21	D
22	D
23	B
24	B
25	B
26	A
27	C
28	B
29	D
30	A
31	C
32	C
33	A
34	A
35	D
36	D
37	E

38	B
39	B
40	B
41	B
42	A
43	C
44	A
45	E
46	E
47	B
48	D
49	E
50	A
51	B
52	B
53	D
54	A
55	D
56	D
57	A
58	D
59	E
60	B
61	D
62	E
63	B
64	E
65	D
66	D
67	B
68	E
69	D
70	D
71	E
72	D
73	A
74	D
75	E

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Question 1



Methylamine, CH_3NH_2 , is a weak base that reacts with water according to the equation above. A student obtains a 50.0 mL sample of a methylamine solution and determines the pH of the solution to be 11.77.

- (a) Write the expression for the equilibrium constant, K_b , for methylamine.

$K_b = \frac{[\text{CH}_3\text{NH}_3^+][\text{OH}^-]}{[\text{CH}_3\text{NH}_2]}$	One point is earned for the correct expression.
--	---

- (b) Calculate the molar concentration of OH^- in the 50.0 mL sample of the methylamine solution.

pH = 11.77 $[\text{H}^+] = 10^{-11.77} = 1.7 \times 10^{-12}$ $[\text{OH}^-] = \frac{K_w}{[\text{H}^+]} = \frac{1.0 \times 10^{-14}}{1.7 \times 10^{-12}} = 5.9 \times 10^{-3}$ OR pOH = 14 – pH = 2.23 $[\text{OH}^-] = 10^{-2.23} = 5.9 \times 10^{-3}$	One point is earned for correct $[\text{OH}^-]$.
---	---

- (c) Calculate the initial molar concentration of $\text{CH}_3\text{NH}_2(aq)$ in the solution before it reacted with water and equilibrium was established.

$K_b = \frac{[\text{CH}_3\text{NH}_3^+][\text{OH}^-]}{[\text{CH}_3\text{NH}_2]}$ $4.4 \times 10^{-4} = \frac{(5.9 \times 10^{-3})(5.9 \times 10^{-3})}{(x - 5.9 \times 10^{-3})}$ $x = \mathbf{0.085\text{ M}}$	One point is earned for $[\text{CH}_3\text{NH}_3^+] = [\text{OH}^-]$. One point is earned for the correct initial molar concentration. <u>Note:</u> An approximated molar concentration does <i>not</i> earn the second point.
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Question 1 (continued)

The 50.0 mL sample of the methylamine solution is titrated with an HCl solution of unknown concentration. The equivalence point of the titration is reached after a volume of 36.0 mL of the HCl solution is added. The pH of the solution at the equivalence point is 5.98.

(d) Write the net-ionic equation that represents the reaction that takes place during the titration.

$\text{CH}_3\text{NH}_2 + \text{H}_3\text{O}^+ \rightarrow \text{CH}_3\text{NH}_3^+ + \text{H}_2\text{O}$ OR $\text{CH}_3\text{NH}_2 + \text{H}^+ \rightarrow \text{CH}_3\text{NH}_3^+$	One point is earned for a correct equation.
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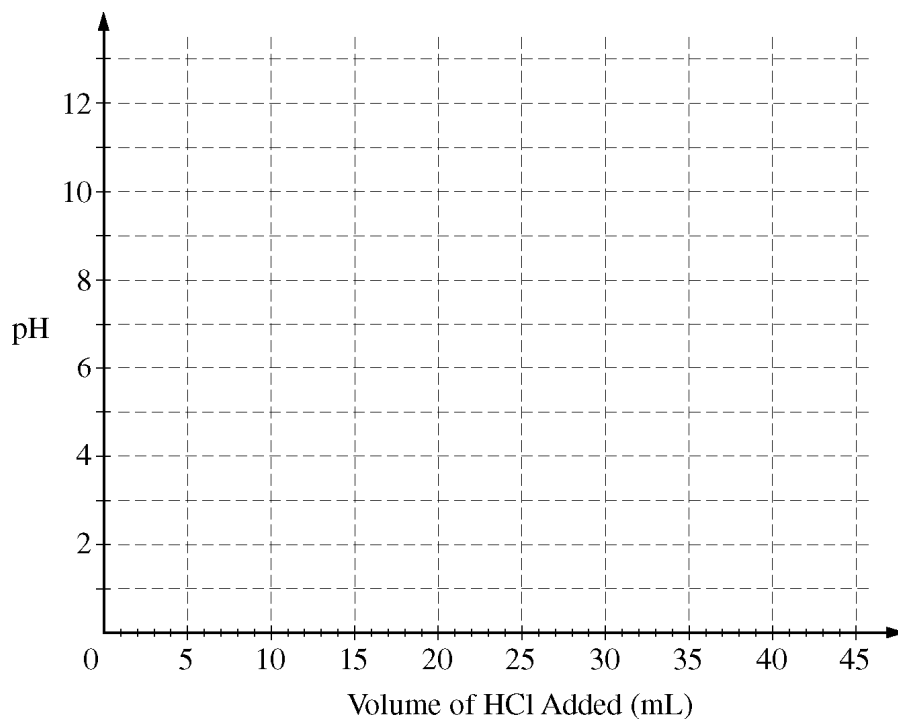
(e) Calculate the concentration of the HCl solution used to titrate the methylamine.

$\frac{0.085 \text{ mol}}{1000. \text{ mL}} \times 50.0 \text{ mL} = 0.00425 \text{ mol CH}_3\text{NH}_2$ $\frac{0.00425 \text{ mol HCl}}{36.0 \text{ mL}} \times \frac{1000. \text{ mL}}{1.000 \text{ L}} = \mathbf{0.12 \text{ M}}$	One point is earned for equal moles of acid and base. One point is earned for the correct concentration.
---	--

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Question 1 (continued)

- (f) Using the axes provided, sketch the titration curve that results from the titration described above. On the graph, clearly label the equivalence point of the titration.



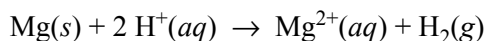
One point is earned for a curve starting at a pH between 11 and 12.

One point is earned for labeling the equivalence point at $V = \sim 36.0$ mL HCl and a pH of ~ 5.98 .

One point is earned for general shape of the curve for a weak acid/strong base titration.

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Question 2



A student performs an experiment to determine the volume of hydrogen gas produced when a given mass of magnesium reacts with excess $\text{HCl}(aq)$, as represented by the net ionic equation above. The student begins with a 0.0360 g sample of pure magnesium and a solution of 2.0 M $\text{HCl}(aq)$.

- (a) Calculate the number of moles of magnesium in the 0.0360 g sample.

$\frac{1 \text{ mol Mg}}{24.30 \text{ g Mg}} \times 0.0360 \text{ g} = 0.00148 \text{ mol Mg}$	One point is earned for the correct number of moles of Mg. <u>Note:</u> For errors in significant figures or mathematics, there is a 1 point deduction; For the entire question, only 1 point total can be deducted for each type of error.
--	---

- (b) Calculate the number of moles of $\text{HCl}(aq)$ needed to react completely with the sample of magnesium.

$0.00148 \text{ mol Mg} \times \frac{2 \text{ mol H}^+}{1 \text{ mol Mg}} = 0.00296 \text{ mol HCl}$	One point is earned for the correct number of moles of HCl.
--	--

As the magnesium reacts, the hydrogen gas produced is collected by water displacement at 23.0°C. The barometric pressure in the lab is measured to be 749 torr.

- (c) Given that the equilibrium vapor pressure of water is 21 torr at 23.0°C, calculate the pressure that the $\text{H}_2(g)$ produced in the reaction would have if it were dry.

$P = 749 \text{ torr} - 21 \text{ torr} = 728 \text{ torr}$	One point is earned for the correct pressure.
---	--

- (d) Calculate the volume, in liters measured at the conditions in the laboratory, that the $\text{H}_2(g)$ produced in the reaction would have if it were dry.

$PV = nRT \Rightarrow V = \frac{nRT}{P}$ $728 \text{ torr} \times \frac{1 \text{ atm}}{760 \text{ torr}} = 0.958 \text{ atm}$ $V = \frac{(0.00148 \text{ mol H}_2)(0.0821 \text{ L atm mol}^{-1} \text{ K}^{-1})(296 \text{ K})}{0.958 \text{ atm}}$ $= 0.0375 \text{ L}$	One point is earned for the correct moles of $\text{H}_2(g)$. One point is earned for agreement of units of P (from part(c)) and R. One point is earned for the correct temperature substitution and calculated volume.
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Question 2 (continued)

- (e) The laboratory procedure specified that the concentration of the HCl solution be 2.0 *M*, but only 12.3 *M* HCl solution was available. Describe the steps for safely preparing 50.0 mL of 2.0 *M* HCl(aq) using 12.3 *M* HCl solution and materials selected from the list below. Show any necessary calculation(s).

10.0 mL graduated cylinder

Distilled water

250 mL beakers

Balance

50.00 mL volumetric flask

Dropper

$$\frac{2.0 \text{ mol HCl}}{1000. \text{ mL}} \times 50.0 \text{ mL} = 0.10 \text{ mol HCl}$$

$$\frac{1000. \text{ mL}}{12.3 \text{ mol HCl}} \times 0.10 \text{ mol HCl} = 8.1 \text{ mL}$$

Pour some distilled water into the volumetric flask so it is about half full. Use the graduated cylinder to measure 8.1 mL of 12.3 *M* HCl and pour it into the volumetric flask. Add a few mL of distilled water to the graduated cylinder, swirl, and add the liquid to the volumetric flask. Swirl the flask to mix. Carefully add more distilled water to the flask, using the dropper for the last few mL to fill the flask to the mark with distilled water.

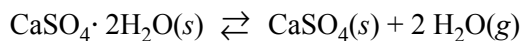
One point is earned for calculating the correct volume of 12.3 *M* HCl.

One point is earned for adding acid to water in the dilution process.

One point is earned for filling the volumetric to the 50.00 mL mark with distilled water.

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Question 3



The hydrate $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}(s)$ can be heated to form the anhydrous salt, $\text{CaSO}_4(s)$, as shown by the reaction represented above.

- (a) Using the data in the table below, calculate the value of ΔG° , in kJ mol^{-1} , for the reaction at 298 K.

Substance	ΔG_f° at 298 K (kJ mol^{-1})
$\text{CaSO}_4 \cdot 2\text{H}_2\text{O}(s)$	-1795.70
$\text{CaSO}_4(s)$	-1320.30
$\text{H}_2\text{O}(g)$	-228.59

$$\Delta G^\circ = \Sigma \Delta G_f^\circ \text{ products} - \Sigma \Delta G_f^\circ \text{ reactants}$$

$$= -1320.30 + (2 \times -228.59) - (-1795.70 \text{ kJ mol}^{-1})$$

$$= \mathbf{18.22 \text{ kJ mol}^{-1}}$$

One point is earned
for the mole factor for water.

One point is earned for answer
(units are optional)

- (b) Given that the value of ΔH° for the reaction at 298 K is $+105 \text{ kJ mol}^{-1}$, calculate the value of ΔS° for the reaction at 298 K. Include units with your answer.

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

$$\Delta S^\circ = \frac{\Delta H^\circ - \Delta G^\circ}{T} = \frac{(105 - 18.22) \text{ kJ mol}^{-1}}{298 \text{ K}}$$

$$= \mathbf{0.29 \text{ kJ K}^{-1} \text{ mol}^{-1} \text{ OR } 290 \text{ J K}^{-1} \text{ mol}^{-1}}$$

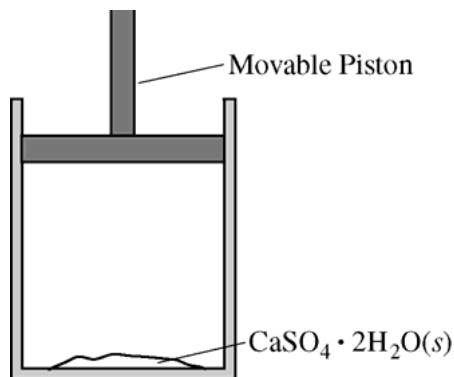
One point is earned for the correct substitution
of ΔG° , ΔH° , and T.

One point is earned for the correct
answer *with correct units*.

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Question 3 (continued)

A sample of $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}(s)$ is placed in a cylinder with a movable piston as shown in the diagram below. The air above the solid is at 1.00 atm and is initially dry (partial pressure of $\text{H}_2\text{O}(g) = 0$ atm).



(c) Write the expression for the equilibrium constant, K_p , for the reaction.

$K_p = (p_{\text{H}_2\text{O}})^2$	One point is earned for the correct expression (use of partial pressure only).
------------------------------------	--

(d) Given that the equilibrium constant, K_p , is 6.4×10^{-4} at 298 K, determine the partial pressure, in atm, of water vapor in the cylinder at equilibrium at 298 K.

$K_p = (p_{\text{H}_2\text{O}})^2 \Rightarrow 6.4 \times 10^{-4}$ $p_{\text{H}_2\text{O}} = \sqrt{6.4 \times 10^{-4}} = \mathbf{0.025 \text{ atm}}$	One point is earned for the correct partial pressure (units are not required).
--	--

(e) If the volume of the system is reduced to one-half of its original volume and the system is allowed to reestablish equilibrium at 298 K, what will be the pressure, in atm, of the water vapor at the new volume? Justify your answer.

The $p_{\text{H}_2\text{O}}$ at equilibrium at the new volume will be 0.025 atm. Equilibrium vapor pressure is dependent on K_p, which in turn is a function of temperature, not volume. Because the temperature is still 298 K, the vapor pressure of H_2O remains 0.025 atm in the new volume.	One point is earned for the correct answer with justification.
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Question 3 (continued)

In the laboratory, the hydrate $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}(s)$ can be heated in a crucible to completely drive off the water of hydration to form the anhydrous salt, $\text{CaSO}_4(s)$.

- (f) A 2.49 g sample of pure $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}(s)$ is heated several times until the mass is constant. Calculate the mass, in grams, of the solid that remains after the dehydration reaction is complete.

$$\text{molar mass of } \text{CaSO}_4 \cdot 2\text{H}_2\text{O}(s) = 172.172 \text{ g mol}^{-1}$$

$$\text{molar mass of } \text{CaSO}_4(s) = 136.14 \text{ g mol}^{-1}$$

$$2.49 \text{ g } \text{CaSO}_4 \cdot 2\text{H}_2\text{O} \times \frac{136.14 \text{ g } \text{CaSO}_4}{172.172 \text{ g } \text{CaSO}_4 \cdot 2\text{H}_2\text{O}} = 1.97 \text{ g } \text{CaSO}_4$$

One point is earned for the (rounded) correct molar masses.

One point is earned for an answer consistent with the molar masses.

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Question 4

For each of the following three reactions, in part (i) write a balanced equation for the reaction and in part (ii) answer the question about the reaction. In part (i), coefficients should be in terms of lowest whole numbers. Assume that solutions are aqueous unless otherwise indicated. Represent substances in solutions as ions if the substances are extensively ionized. Omit formulas for any ions or molecules that are unchanged by the reaction. You may use the empty space at the bottom of the next page for scratch work, but only equations that are written in the answer boxes provided will be graded.

(a) Solutions of ethanoic (acetic) acid and lithium hydroxide are combined.

(i) Balanced equation: $\text{HC}_2\text{H}_3\text{O}_2 + \text{OH}^- \rightarrow \text{C}_2\text{H}_3\text{O}_2^- + \text{H}_2\text{O}$	One point is earned for the correct reactants. Two points are earned for the correct products. One point is earned for balancing the equation for mass and charge.
---	---

(ii) If the ethanoic acid is titrated with the lithium hydroxide, is the pH at the equivalence point equal to 7, less than 7, or greater than 7? Explain.

The pH will be greater than 7. $\text{HC}_2\text{H}_3\text{O}_2$ is a weak acid and LiOH is a strong base. At the equivalence point, the $\text{C}_2\text{H}_3\text{O}_2^-$ is the pH-determining species because of the reaction, $\text{C}_2\text{H}_3\text{O}_2^- + \text{H}_2\text{O} \rightleftharpoons \text{HC}_2\text{H}_3\text{O}_2 + \text{OH}^-$.	One point is earned for the correct answer with an explanation.
--	--

(b) Solutions of iron(III) chloride and tin(II) chloride are combined.

(i) Balanced equation: $2 \text{Fe}^{3+} + \text{Sn}^{2+} \rightarrow 2 \text{Fe}^{2+} + \text{Sn}^{4+}$ OR $2 \text{Fe}^{3+} + 3 \text{Sn}^{2+} \rightarrow 2 \text{Fe} + 3 \text{Sn}^{4+}$	One point is earned for the correct reactants. Two points are earned for the correct products. One point is earned for balancing the equation for both mass and charge.
---	--

(ii) Identify the species that is reduced in the reaction.

Fe^{3+} is reduced.	One point is earned for correct answer.
--	--

AP[®] CHEMISTRY 2012 SCORING GUIDELINES (INTERNATIONAL)**Question 4 (continued)**

(c) Solutions of silver(I) nitrate and sodium phosphate are combined, forming a precipitate.

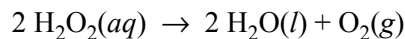
(i) Balanced equation: $3 \text{Ag}^+ + \text{PO}_4^{3-} \rightarrow \text{Ag}_3\text{PO}_4$	Two points are earned for the correct reactants. One point is earned for the correct product. One point is earned for balancing the equation for both mass and charge.
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(ii) If 5 mol of silver(I) nitrate and 3 mol of sodium phosphate react as completely as possible, which reactant, if any, is present in excess? Justify your answer.

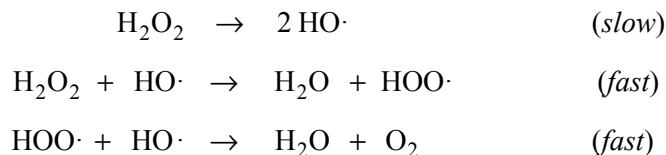
Nine moles of Ag^+ would be needed to react completely with the three moles of PO_4^{3-}, so sodium phosphate is present in excess.	One point is earned for a correct answer with reference to reactant data.
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Question 5



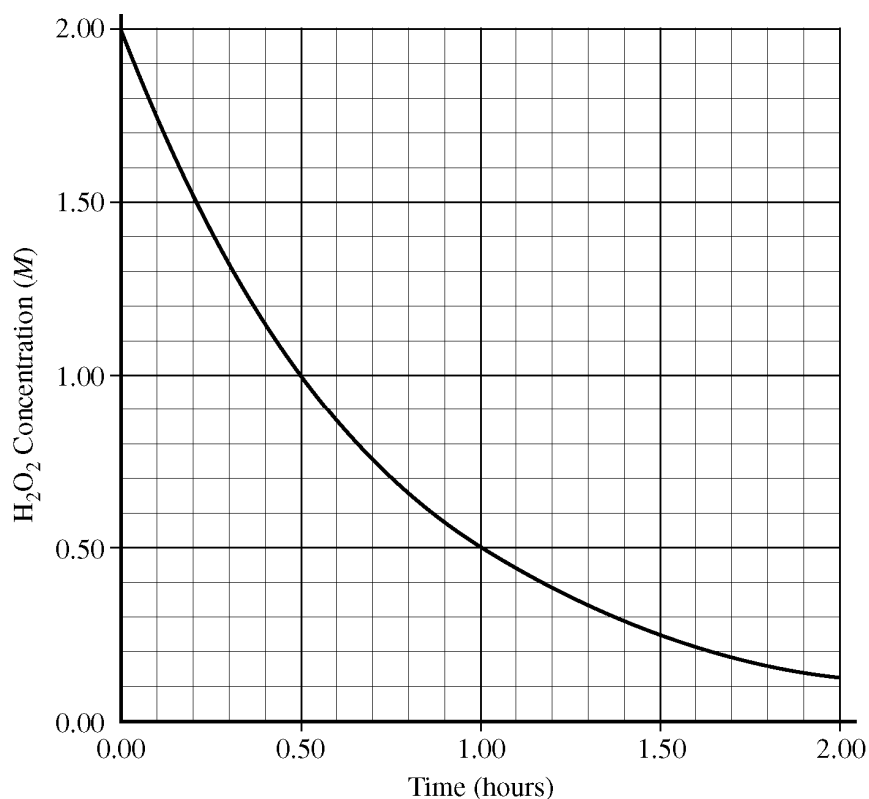
The decomposition of hydrogen peroxide to form water and oxygen gas is represented by the equation above. A proposed mechanism for the reaction, which involves the free radicals $\text{HO}\cdot$ and $\text{HOO}\cdot$, is represented by the three equations below.



- (a) Write the rate law consistent with the proposed mechanism above.

$\text{rate} = k [\text{H}_2\text{O}_2]$	One point is earned for the correct rate law.
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- (b) The rate of the decomposition reaction was studied in an experiment, and the resulting data were plotted in the graph below.



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Question 5 (continued)

Using the graph, determine the time, in hours, needed for the concentration of H_2O_2 to change from

(i) 1.50 *M* to 0.75 *M*

Time = $0.71 - 0.21 = 0.50 \pm 0.05$ hour	One point is earned for the correct time.
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(ii) 0.80 *M* to 0.40 *M*

Time = $1.17 - 0.66 = 0.51 \pm 0.05$ hour	One point is earned for the correct time.
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(c) The experimental data are consistent with the proposed mechanism. Explain.

The data show that the time remains constant when the concentration of the reactant decreases by one-half, indicating that the reaction is first order. The rate law for the proposed mechanism has $[\text{H}_2\text{O}_2]$ to the first power, indicating a first-order reaction.	One point is earned for a correct explanation.
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An electrochemical cell based on the decomposition of H_2O_2 can be constructed based on the half-reactions in the table below.

Half-Reaction	Standard Reduction Potential, E°
$\text{H}_2\text{O}_2 + 2 e^- \rightarrow 2 \text{OH}^-$	0.88 V
$\text{O}_2 + 2 \text{H}_2\text{O} + 2 e^- \rightarrow \text{H}_2\text{O}_2 + 2 \text{OH}^-$	-0.15 V

(d) Calculate the value of the standard cell potential, E° , for the cell.

$\text{H}_2\text{O}_2 + 2 \text{OH}^- \rightarrow \text{O}_2 + 2 \text{H}_2\text{O} + 2 e^-$ 0.15 V $\text{H}_2\text{O}_2 + 2 e^- \rightarrow 2 \text{OH}^-$ 0.88 V $2 \text{H}_2\text{O}_2 \rightarrow 2 \text{H}_2\text{O} + \text{O}_2$ 1.03 V	One point is earned for the correct value of E° .
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Question 5 (continued)

- (e) Indicate whether ΔG° for the decomposition reaction is greater than 0, less than 0, or equal to 0. Justify your answer.

ΔG° is less than 0. $\Delta G^\circ = -nFE^\circ$. Because E° is positive, ΔG° must be negative. OR E° is positive, thus the reaction is spontaneous. For spontaneous reactions, ΔG° is always negative.	One point is earned for the correct answer with justification.
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- (f) The decomposition of $\text{H}_2\text{O}_2(aq)$ is slow at 298 K, but a suitable catalyst greatly increases the rate of the decomposition reaction.

- (i) Draw a circle around each of the quantities below that has a different value for the catalyzed reaction than for the uncatalyzed reaction.

K_{eq}

ΔG°

ΔH°

E_a

Only E_a should be circled.	One point is earned for circling E_a and none of the other quantities.
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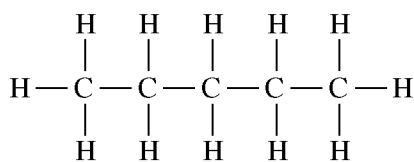
- (ii) For any quantity that you circled above, indicate whether its value is greater or less for the catalyzed reaction than for the uncatalyzed reaction. Explain why.

E_a , the activation energy, is less for the catalyzed reaction than for the uncatalyzed reaction. A catalyst provides a new reaction mechanism that requires a lower minimum required potential energy for the reaction to occur. Thus the activation energy is lowered. <u>Note:</u> reference to other circled quantities is ignored.	One point is earned for a correct answer with reference to an altered mechanism.
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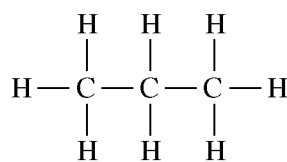
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Question 6

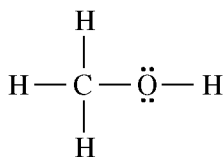
Answer the following questions in terms of principles of chemical bonding and intermolecular forces. In each explanation where a comparison is to be made, a complete answer must include a discussion of both substances. The following complete Lewis electron-dot diagrams may be useful in answering parts of this question.



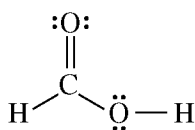
Pentane



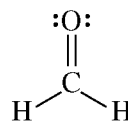
Propane



Methanol



Methanoic (formic) acid



Methanal (formaldehyde)

(a) At 1 atm and 298 K, pentane is a liquid whereas propane is a gas. Explain.

Both molecules are nonpolar, and the only intermolecular forces in each are London dispersion forces. However, pentane is larger than propane and has a more extensive electron cloud that can be involved in a greater number of London interactions, leading to stronger intermolecular attractions overall. Thus it takes a higher temperature for pentane molecules to have enough kinetic energy (on average) to overcome their stronger intermolecular attractions, thus pentane has the higher boiling point.

One point is earned for recognizing that pentane and propane *both* have only LDF's.
One point is earned for recognizing that pentane has a larger electron cloud with greater IMF's and higher BP (reference to state of matter).

(b) At 1 atm and 298 K, methanol is a liquid whereas propane is a gas. Explain.

Propane molecules are nonpolar and only interact with one another via London dispersion forces. Methanol molecules are polar and hydrogen bonds (as well as London forces) can form among them. Because hydrogen bonds are stronger than London forces, methanol has greater intermolecular attractions. Thus it takes a higher temperature for methanol molecules to have enough kinetic energy (on average) to overcome their stronger intermolecular attractions, thus methanol has the higher boiling point.

One point is earned for recognizing that methanol has hydrogen-bonding IMF's and propane has LDF's.

One point is earned for recognizing that hydrogen-bonding results in a greater IMF causing methanol to be a liquid.

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Question 6 (continued)

(c) Indicate the hybridization of the carbon atom in each of the following:

(i) Methanol

sp^3	One point is earned for the correct hybridization.
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(ii) Methanoic (formic) acid

sp^2	One point is earned for the correct hybridization.
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(d) Draw the complete Lewis electron-dot diagram for a molecule of propanoic acid, $\text{HC}_3\text{H}_5\text{O}_2$.

$ \begin{array}{ccccccc} & \text{H} & & \text{H} & & :\text{O}: & \\ & & & & & & \\ \text{H} & -\text{C} & - & \text{C} & - & \text{C} & - \ddot{\text{O}} - \text{H} \\ & & & & & \ddot{\cdot} & \\ & \text{H} & & \text{H} & & & \end{array} $	One point is earned for the correct diagram (all electron pairs must be included).
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(e) Explain the following observations about the two carbon-oxygen bonds in the methanoate (formate) anion, HCO_2^- . You may draw a Lewis electron-dot diagram (or diagrams) of the methanoate ion as part of your explanations.

(i) The two carbon-oxygen bonds in the methanoate (formate) anion, HCO_2^- , have the same length.

Resonance structures can be drawn for the methanoate anion, with one carbon-oxygen bond as a single bond and the other as a double bond (and the opposite in the other structure). The electrons are shared equally between the two carbon-oxygen bonds in the methanoate anion, resulting in two bonds with the same length.	One point is earned for the correct explanation.
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(ii) The length of the carbon-oxygen bonds in the methanoate (formate) anion, HCO_2^- , is intermediate between the length of the carbon-oxygen bond in methanol and the length of the carbon-oxygen bond in methanal.

The two identical carbon-oxygen bonds in the methanoate anion each have a bond order of 1.5. The carbon-oxygen bond in methanol is a single bond, and the carbon-oxygen bond in methanal is a double bond. Single bonds between atoms of the same two elements are longer than double bonds, and a bond with a bond order of 1.5 would have a bond length between that of a single bond and that of a double bond.	One point is earned for the correct explanation comparing the single, double <i>and</i> resonating bonds of the respective molecules.
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